

Stony Brook University
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Spring 2017

EXTRA CREDIT ASSIGNMENT

Due: Optional Tuesday, May 9 before the midterm.

INSTRUCTIONS: This is an empirical optional assignment for extra credit purposes. Please write your answers as comments in an R script and deliver a printout of the code together with the responses. **Note:** For all your hypothesis tests and confidence intervals, you must use **robust standard errors**.

1. Use the data in WAGE1 for this problem.

(a) Use OLS to estimate the equation

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{exper}^2 + u$$

and report the results using the usual format.

(b) Is exper^2 statistically significant at the 1% level?

(c) Using the approximation

$$\% \Delta \widehat{\text{wage}} \approx 100(\hat{\beta}_2 + 2\hat{\beta}_3 \text{exper}) \Delta \text{exper},$$

find the approximate return to the fifth year of experience. What is the approximate return to the twentieth year of experience? Put a 95% confidence interval around the predicted returns at both the fifth and twentieth year of experience.

(d) At what value of exper does additional experience actually lower predicted $\log(\text{wage})$? How many people have more experience in this sample?

2. Use the data in VOTE1 for this exercise.

(a) Consider a model with an interaction between expenditures:

$$\text{voteA} = \beta_0 + \beta_1 \text{prtystrA} + \beta_2 \text{expendA} + \beta_3 \text{expendB} + \beta_4 \text{expendA} \times \text{expendB} + u$$

What is the partial effect of expendB on voteB , holding prtystrA and expendA fixed? What is the partial effect of expendA on voteA ? Is the expected sign for β_4 obvious?

(b) Estimate the equation in part (a) and report the results in the usual form. Is the interaction term statistically significant?

(c) Find the average of expendA in the sample. Fix expendA at 300 (for \$300,000). What is the estimated effect of another \$100,000 spent by Candidate B on voteA ? Is this a large effect?

(d) Now fix expendB at \$100. What is the estimated effect of $\Delta \text{expend} = 100$ on voteA ? Does this make sense? Estimate a 99% confidence interval around the predicted change on voteA .

(e) Now, estimate a model that replaces the interaction with shareA , candidate A's percentage share of total campaign expenditures. Does it make sense to hold both expendA and expendB fixed, while changing shareA ? How would you interpret the coefficient on shareA ? *Hint: Notice that $\text{shareA} = 100 \frac{\text{expendA}}{\text{expendA} + \text{expendB}}$ so when you change, say expendA , shareA also moves, and similarly when you change expendB .*

- (f) In the model from part (e), find the partial effect of $expendB$ on $voteA$, holding $prtystrA$ and $expendA$ fixed. Evaluate this at $expendA = 300$ and $expendB = 0$ and compute a 95 confidence interval around this partial effect. Comment on the results. *Hint: Notice that $shareA = 100 \frac{expendA}{expendA + expendB}$, so when computing the partial effect¹ of $expendB$ you will need to take this into account.*
3. The data set NBASAL contains salary information and career statistics for 269 players in the National Basketball Association (NBA).
- Estimate a model relating points-per-game ($points$) to years in the league ($exper$), age , and years played in college ($coll$). Include a quadratic in $exper$; the other variables should appear in level form. Report the results in the usual way.
 - Holding college years and age fixed, at what value of experience does the next year of experience actually reduce points-per-game? Does this makes sense?
 - Why do you think $coll$ has a negative and statistically significant coefficient? (*Hint: NBA players can be drafted before finishing their college careers and even directly out of high school.*²)
 - Add a quadratic in age to the equation. Is it needed? What does this appear to imply about the effects of age, once experience and education are controlled for?
 - Now regress $\log(wage)$ on $points$, $exper$, $exper^2$, age and $coll$. Report the results in the usual format.
 - Test whether age and $coll$ are jointly significant in the regression from part (e). What does this imply about whether age and education have separate effects on wage, once productivity and seniority are accounted for?

¹Partial effect means partial derivative.

²Like was the case for Kobe Bryant.